SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: CSA16 COURSE NAME: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3, BL4, BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3 **Total Marks:30**

Annexure C

Case study

Criteria	Understanding of Case *(2.5)*	Application of Concepts *(2.5)*	Analysis *(2)*	Solution Design *(2)*	Clarity *(1)*	***Total (10)***
Weightage	*25%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						

Code of Conduct:

I Haresh Kumar N L (192425009) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A

Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock but then get mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP, FP, FN, TN

2.Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP, FP, FN, TN and Calculate the Precision and Recall and F1 Score.

3.An email service provider wants to classify emails as **Spam or Not Spam** using features:

- Contains "Offer" (Yes/No)
- Contains "Free" (Yes/No)
- Email Length (Short/Long)

Sample Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:
    - P(Spam), P (Not Spam)
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
    - (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes

Answers:

1) **Shepherd Boy – TP, FP, FN, TN**

The situation is:

- The boy cries **“Wolf!” when there is no wolf** → False Positive.
- Later, a **real wolf appears and the boy cries “Wolf!”** → True Positive.
- There is no example of a wolf being present but the boy not detecting it.
- There is no clear example of correctly reporting no wolf.

Actual Situation	**Prediction**	**Result**
No wolf	Wolf	**FP**
Wolf	Wolf	**TP**
Wolf	No Wolf	**FN**
No wolf	No Wolf	**TN**

- **TP = 1**
- **FP = 1**
- **FN = 0**
- **TN = 0**

The first false alarm is FP because the boy predicted a wolf when there was none. The later genuine alarm is TP because a wolf was actually present.

2) **Cat Image Classification**

Given:

- Total images = **100**
- Cat images = **30**
- No-cat images = **70**
- Model predicts cat correctly for **25 of 30 cat images**
- Model predicts no cat correctly for **50 of 70 no-cat images**

Step 1: Find TP

25 cat images were correctly predicted as cat.

TP = 25

Step 2: Find FN

There are 30 actual cat images.

$FN=30-25=5$

FN = 5

Step 3: Find TN

50 no-cat images were correctly predicted as no cat.

TN = 50

Step 4: Find FP

There are 70 actual no-cat images.

FP=70−50=20

FP = 20

	Actual Cat	**Actual No Cat**
Predicted Cat	TP = 25	FP = 20
Predicted No Cat	FN = 5	TN = 50

Precision:

Precision = (TP/TP + FP)

= (25/25 + 20)

= 25/45

= 0.5556

Precision = 55.56%

Recall:

Recall = (TP/TP + FN)

= (25/25 + 5)

= 25/30

= 0.8333

Recall = 83.33%

F1 Score:

F1 = 2 × (Precision × Recall /Precision + Recall)

= 2 × (0.5556 × 0.8333/0.5556 + 0.5556)

F1 ≈ 0.6667

F1 Score = 66.67%

3) Naïve Bayes – Spam Classification

Given dataset contains 6 emails with the features Offer, Free, Length and Class:

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

I) Prior Probabilities

There are:

- **Spam = 3**
- **Not Spam = 3**
- **Total = 6**

Therefore,

P(Spam) = 3/6

= 0.5

P(NotSpam) = 3/6

= 0.5

II) Conditional Probabilities

For Spam

There are 3 Spam emails.

Offer

P(Offer=Yes|Spam) = 2/3

P(Offer=No|Spam) = 1/3

Free

P(Free=Yes|Spam) = 2/3

P(Free=No|Spam) = 1/3

Length

P(Length=Short|Spam) = 2/3

P(Length=Long|Spam) = 1/3

For Not Spam

There are 3 Not Spam emails.

Offer

$P(Offer=Yes|NotSpam) = 1/3$

$P(Offer=No|NotSpam) = 2/3$

Free

$P(Free=Yes|NotSpam) = 0/3 = 0$

$P(Free=No|NotSpam) = 3/3 = 1$

Length

$P(Length=Short|NotSpam) = 1/3$

$P(Length=Long|NotSpam) = 2/3$

III) Classify: Offer = Yes, Free = No, Length = Short

According to the question, we need to classify the email using Naïve Bayes.

Probability for Spam:

$P(Spam|X) \propto P(Spam) \times P(Offer=Yes|Spam) \times P(Free=No|Spam) \times P(Short|Spam)$

$= (1/2) \times (2/3) \times (1/3) \times (2/3)$

$= 2/27$

$= 0.07407$

Probability for Not Spam:

$P(NotSpam|X) \propto P(NotSpam) \times P(Offer=Yes|NotSpam) \times P(Free=No|NotSpam) \times P(Short|NotSpam)$

$= (1/2) \times (1/3) \times (1) \times (1/3)$

$= 1/18$

$= 0.05556$

Since:

$0.07407 > 0.05556$

Classification:

The email is classified as SPAM.

Normalized probabilities:

P(Spam|X)≈57.14%

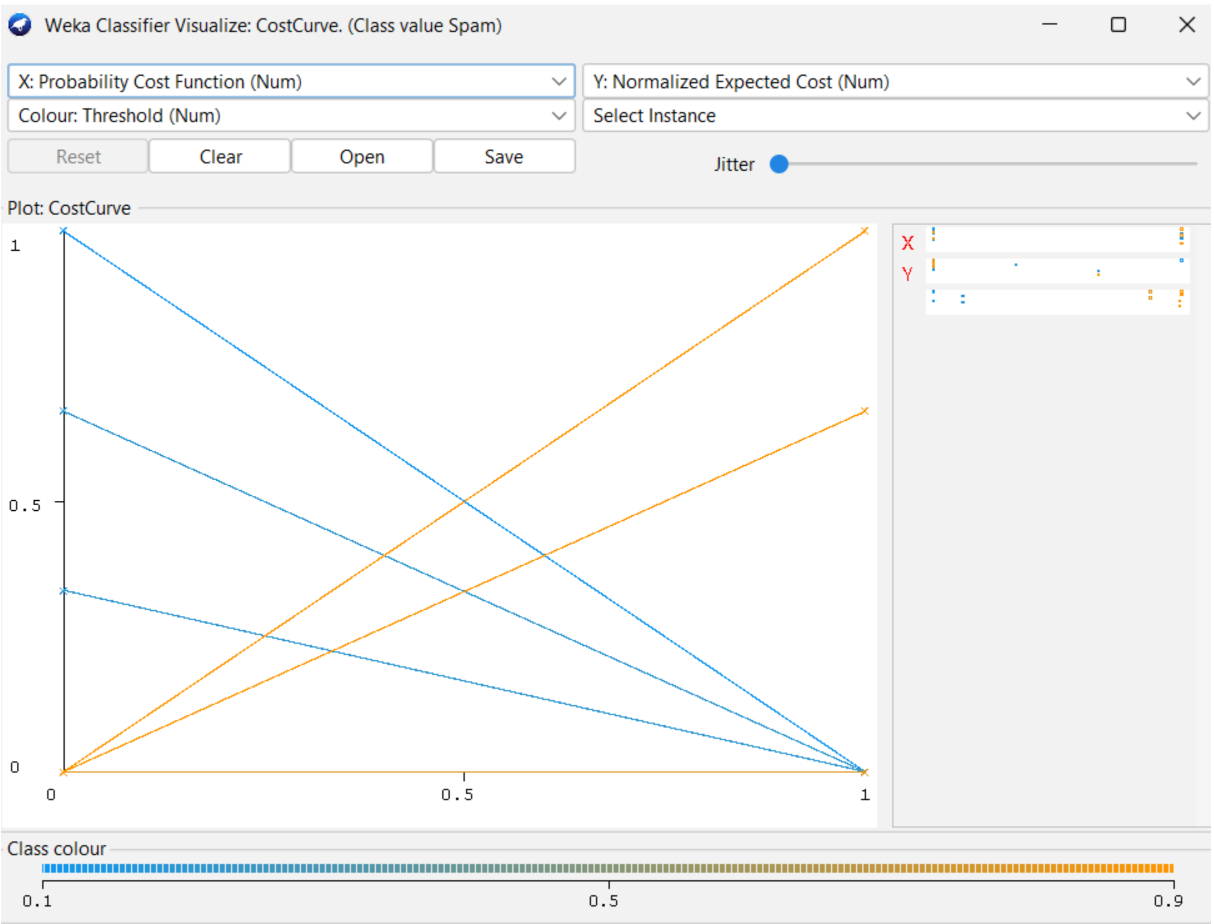
P(NotSpam|X)≈42.86%

VI)

Classifier output

```
=== Run information ===

Scheme:       weka.classifiers.bayes.NaiveBayes
Relation:     email_spam
Instances:    6
Attributes:   4
              Offer
              Free
              Length
              Class
Test mode:    evaluate on training data

=== Classifier model (full training set) ===

Naive Bayes Classifier

                  Class
Attribute          Spam Not Spam
                  (0.5)    (0.5)
=================================
Offer
  Yes               3.0      2.0
  No                2.0      3.0
  [total]           5.0      5.0

Free
  Yes               4.0      1.0
  No                1.0      4.0
  [total]           5.0      5.0

Length
  Short             3.0      2.0
  Long              2.0      3.0
  [total]           5.0      5.0



Time taken to build model: 0 seconds

=== Evaluation on training set ===
```

```
Time taken to test model on training data: 0 seconds

=== Summary ===

Correctly Classified Instances           6              100      %
Incorrectly Classified Instances         0                0      %
Kappa statistic                          1
Mean absolute error                      0.1667
Root mean squared error                  0.1732
Relative absolute error                 33.3333 %
Root relative squared error             34.641  %
Total Number of Instances                6

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 1.000    0.000    1.000      1.000    1.000      1.000    1.000     1.000     Spam
                 1.000    0.000    1.000      1.000    1.000      1.000    1.000     1.000     Not Spam
Weighted Avg.    1.000    0.000    1.000      1.000    1.000      1.000    1.000     1.000

=== Confusion Matrix ===

 a b   <-- classified as
 3 0 | a = Spam
 0 3 | b = Not Spam
```



V) Independence Assumption in Naïve Bayes

Naïve Bayes assumes that the features are conditionally independent given the class.

In this case, it assumes that:

- **Offer**
- **Free**

- **Email Length**

are independent of each other when we already know whether the email is Spam or Not Spam.

For example, Naïve Bayes calculates:

$$P(Spam|X) \propto P(Spam)P(Offer|Spam)P(Free|Spam)P(Length|Spam)$$

Advantages

- **Simple and fast.**
- **Works well with small datasets.**
- **Easy to implement.**
- **Requires relatively little training data.**

Limitation

In real email data, features may not actually be independent. For example, an email containing “Free” may also be more likely to contain “Offer”. Therefore, the independence assumption may not always hold perfectly.